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Numerical study of internal dynamics and mass transfer distribution of optical fiber laser offset welding of titanium and aluminum

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ABSTRACT

The welding of different metals will produce intermetallic compounds, and the mixing of molten phases and the diffusion of elements in the welding process will affect the quality of welded joints to varying degrees. In this study, the internal mechanism that affects the joint quality during optical fibers laser offset welding of different metal was explored, and a three-dimensional transient numerical model was established. The heat transfer, fluid flow, mass transfer distribution and the evolution of keyhole surface between titanium and aluminum were studied. The adaptive heat source model is used to track the shape change inside the keyhole, and the surface shape and stress of the keyhole are calculated by the fluid volume method. The influences of steam shear stress, surface tension and recoil pressure on the molten pool are mainly considered. The change in welding power, welding speed and laser offset will cause a change in the temperature field in the molten pool and the thermal convection of liquid metal. The mechanism of heat transfer, flow and driving force in the molten pool is analyzed by dimensionless width. The results show that the change in laser power and laser speed has a significant influence on the shape change in the molten pool. When the molten pool reaches a quasi-steady state, a dynamic balance will be formed between the recoil pressure and the surface tension of the molten pool. As the main driving force, Marangoni force controls the change in the molten pool, and the change in welding parameters will affect the welding quality by producing intermetallic transition layers with different thicknesses. The simulation results are in good agreement with the experimental results in the literature. These understandings can provide new insights for different metal offset welding and provide guidance for optimizing manufacturing process parameters.

1. Introduction

Due to the rapid development of laser welding technology, its advantages have become more obvious. Precise welding positioning and high energy utilization make laser welding technology widely used in many industries such as automotive, oil and gas, shipbuilding, aerospace industry, etc.[1]. Compared with other joining methods such as riveting and screwing, the laser has the unique advantages of welding, which can control the size of energy input, precise positioning, high energy utilization, and good penetration performance of metal.

Modern industries such as aerospace, automotive, oil and gas require superior performance materials to meet the demands of applications for different environments. These materials need to have a high degree of suitability, such as lightweight and wear resistance. It has been found that the performance of a single material is often not as good as a mixture of two or more materials and that multiple combinations of materials are required to meet the needs of these industries [2]. Aluminum alloys have a lower density, melting point, better corrosion resistance, strength and thermal conductivity. Titanium alloy has similar properties to aluminum alloy, and it has good biocompatibility and high heat resistance. Titanium alloys and aluminum alloys are both lightweight metals, and titanium-aluminum lightweight hybrid structures have a wide range of applications and potential development prospects in the aerospace, aircraft and automotive industries [3].

Different thermophysical properties (specific heat, melting point, etc.) of titanium and aluminum materials make the connection between them difficult in many ways. During the welding process, the irregular growth of metal phases and the constant mixing of melting phases tend

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to form brittle intermetallic compounds ($TiAl_3$, $TiAl_2$, Ti_3Al), which greatly weaken the mechanical properties of titanium and aluminum materials. The presence of high thermal stresses within the weld pool, the imbalance of thermal stresses, and the generation of brittle compounds make the internal quality of the weld uneven and prone to cracks. Therefore, it is important to control the laser energy input as well as the laser deflection during the welding process. Controlling laser energy input and keeping the intermetallic transition layer within a reasonable range can effectively reduce the formation of TiAl3-like compounds and can improve the quality of the welded joint [4].

In recent years, researchers have gradually deepened their studies on the internal force behavior of the keyhole, the dynamics of the melt pool, the mutual transfer and convection generation of solutes, the size of the porosity, etc. Pang et al. established a beam-tracking model to simulate the force conditions of keyhole formation, and the results showed that the shape and surface evolution of the keyhole is influenced by the surface tension of the melt pool, the metal vapor shear stress, and the recoil pressure [5,6]. Zhang et al. analyzed the flow characteristics of the weld pool and the mechanism of metal spatter formation, the results show that the shear force and recoil pressure caused by the evaporation of metal vapor is an important factor in the formation of spatter [7]. Hu et al. established a three-dimensional model of stainless steel-nickel heterogeneous joints, analyzed the effect of fluid flow on the temperature field and its evolution, and found that fluid convection has a significant effect on mass transfer [8]. Zhang et al. simulated the dynamic coupling behavior inside the molten pool under full penetration laser welding and analyzed the evolution of the lower part of the molten pool [9]. Akbari et al. predicted the internal geometry of the weld at different welding speeds by analyzing the temperature distribution inside the weld [10]. Casalino et al. analyzed the temperature distribution in the molten pool and thermal cycling by finite element simulation under different welding conditions and joint morphology [11]. Liu et al. simulated the microstructural changes during solidification in the weld pool and predicted the shape as well as the size of the formed grains [12]. In the process of butt welding of titanium and aluminum materials, different laser offset, welding speed, offset position, and energy input has different effects on the mutual coupling of the keyhole and molten pool. Song et al. found that as the laser offset increased, convection at the interface decreased, and the interface temperature, as well as the IMC layer thickness decreased [13]. Tomashchuk et al. showed that when the offset side is aluminum material, the cooling gradient is larger, which makes the interdiffusion as well as mixing inside the melt pool less [14]. Lee et al. showed that the growth of metallurgical organization within the intermetallic transition layer was suppressed at very high laser welding power as well as welding speed [15], This is similar to the findings of Tomashchuk et al. [16]. Mohan et al. analyzed the convection and heat transfer in welding with or without shock by Peclet number, and established the dominant position of surface tension as the driving force in the process of molten pool change [17]. Duggirala et al. Through the analyzed of the dimensionless width of the molten pool, the Marangoni convection in materials with negative gradient of surface tension was explained [18].

In this paper, a three-dimensional model with titanium alloy and aluminum alloy as the butt metal was established through finite element analysis to simulate the effect of different welding conditions on the molten pool flow under the action of fiber laser offset welding. The effects of recoil pressure, surface tension and Marangoni force generated during the liquefaction and vaporization of solid metal on the mutual heat convection of liquid metal were considered. By comparing the relationship between dimensionless width and dimensionless number, the heat transfer and convection in the molten pool were analyzed. The temperature field and thermal circulation inside the melt pool were analyzed and the thermal diffusion phenomenon between the two metals was investigated.

2. Numerical procedure

Butt offset welding of dissimilar metals involves many physical phenomena such as heat conduction, heat convection, spatter, etc. To improve the calculation efficiency, the length and width of the metal material are appropriately reduced, while the thickness is the same as the welded sample. Fig. 1 shows the welding schematic of titanium alloy and aluminum alloy under laser offset welding. We introduce the following assumptions:

(1) The fluid in this model uses the Newtonian viscosity model, and the melt flow is a laminar flow, which is incompressible [19].
(2) The effect of shielding gas on the molten pool as well as the keyhole formation is neglected.
(3) Only titanium and aluminum elements are considered, ignoring the influence of other elements in the welding and not considering the material loss caused by the evaporation of metal vapor.
(4) The two metal sheets are continuous with each other, ignoring the gap between them.
(5) The power density of the laser beam is Gaussian distributed.

2.1. Governing equations

Based on the above assumptions, the equations for conservation of mass, momentum and energy are given as:

$$\frac{\partial \rho}{\partial t}+\nabla\cdot\left(\rho\vec{V}\right)=\frac{\partial(\rho u)}{\partial x}+\frac{\partial(\rho v)}{\partial y}+\frac{\partial(\rho w)}{\partial z}+S_u \tag{1}$$

$$\frac{\partial}{\partial t}\left(\rho\vec{V}\right)+\nabla\cdot\left(\rho\vec{V}\cdot\vec{V}\right)=\nabla\cdot\left(\mu\nabla\vec{V}\right)-\nabla P+\rho\vec{g}+\vec{F_f}+S_v \tag{2}$$

$$\frac{\partial}{\partial t}(\rho H)+\nabla\cdot\left(\rho\vec{V}H\right)=\nabla\cdot(K\nabla T)+S_w \tag{3}$$

where ρ, μ, g, t, P, T, K and V denote material density, dynamic viscosity, gravitational coefficient, time, static pressure, temperature, thermal conductivity, and velocity, respectively. u, v, and w are the velocity components in the x, y, and z directions, respectively. S_u, S_v, and S_w are the mass source, momentum source, and volumetric heat source, respectively, and the buoyancy force $\vec{F_f}$ (Boussinesq approximation) can be expressed as:

$$\vec{F_f}=\rho\vec{g}\beta(T-T_0) \tag{4}$$

β is the coefficient of thermal expansion and H is the enthalpy, which can be expressed as:

$$H = h_{ref}+\int_{T_{ref}}^{T}C_p dT+fL \tag{5}$$

where h_{ref}, T_{ref}, L, C_p and f are reference enthalpy, reference temperature, latent heat of the material, specific heat, and liquid fraction respectively.

In the laser welding process, the keyhole is driven by the recoil pressure, which forms the free surface of the keyhole, and the surface tension prevents the formation of the keyhole, and the recoil pressure P (T) can be expressed as:

$$P(T) = 0.54P_{sat}(T)=0.54P_0\exp(Lv(T-T_0)/(RTT_b)) \tag{6}$$

where P_{sat}, T, P_0, T_b and Lv denote the saturation pressure, temperature, atmospheric pressure, boiling point at atmospheric pressure, and latent heat of evaporation, respectively. The surface tension P_σ can be expressed as:

$$P_\sigma=K_\sigma\gamma_\sigma \tag{7}$$

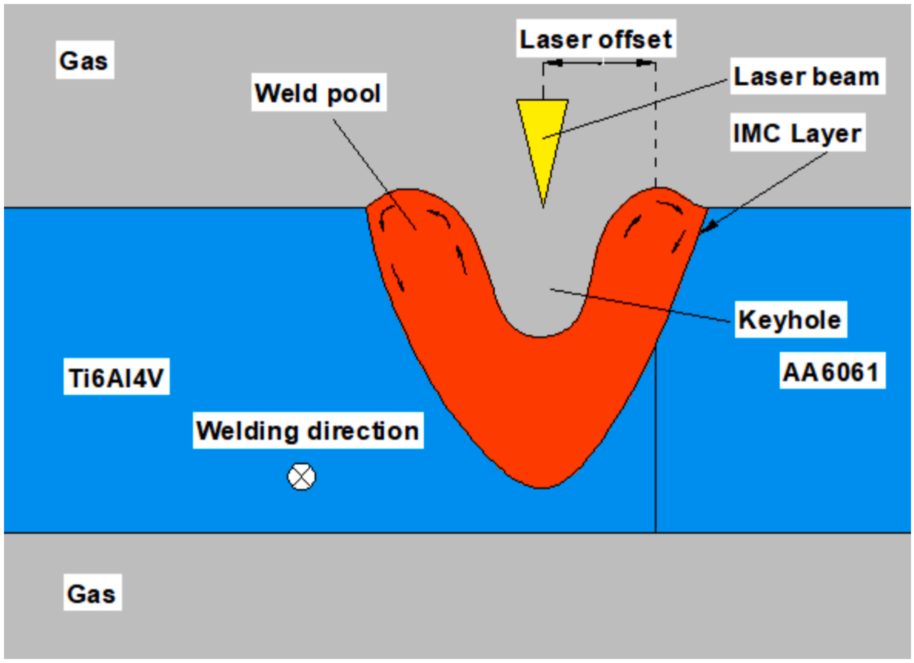

Fig. 1. Schematic diagram of welding Ti6Al4V titanium alloy and 6061 aluminum alloy under laser offset welding.

$$\gamma_\sigma = \gamma_0 - \frac{d\gamma_\sigma}{dT}(T - T_m) \tag{8}$$

where T_m, K_σ and γ_σ represent the melting temperature, the curvature of the surface, and the surface tension coefficient of the workpiece, respectively. Predicting the mass fraction of species can be done using convection and diffusion equations, and the species transport equation can be expressed as:

$$\frac{\partial}{\partial t}(\rho\gamma_i) + \nabla\cdot\left(\rho\vec{V}\gamma_i\right) = -\nabla\vec{J_t} \tag{9}$$

where i, γ_i and J_i denote the species, the mass fraction of species i , and the diffusion flux of species i. and where the diffusion flux J_i can be expressed by:

$$\vec{J_t} = -\rho D_{im}\nabla\gamma_i - D_{Ti}\frac{\nabla T}{T} \tag{10}$$

where D_{im} and D_{Ti} are the mass diffusion coefficient and the thermal diffusion coefficient of species i in the middle, respectively.

The free surface of the keyhole is captured by the volume of fluid method (VOF), and the equation for the volume fraction (F) can be expressed as [20]:

$$\frac{\partial F}{\partial t} + \vec{V}\cdot\nabla F = 0 \tag{11}$$

The gas–liquid changes during the welding process can be expressed in terms of S_{lu} and S_{ul}:

$$S_{lu} = -m_{lu} \tag{12}$$

$$S_{ul} = m_{ul} \tag{13}$$

where S_{lu} and S_{ul} are the mass sources in equation (1), and m_{lu} represents the grid mass, whose positive values represent the heat given off and negative values represent the heat absorbed. The mass distribution of the different elements in the melt pool was calculated by solving the species transport equation (9). The average dynamic viscosity μ, specific heat C_p, density ρ and thermal conductivity k of the Ti/Al mixture can be summarized as [21]:

$$\mu = \gamma_i\mu_1 + (1-\gamma_i)\mu_2 \tag{14}$$

$$C_P = \gamma_i C_{P1} + (1-\gamma_i)C_{P2} \tag{15}$$

$$\rho = \gamma_i\rho_1 + (1-\gamma_i)\rho_2 \tag{16}$$

$$k = \gamma_i k_1 + (1-\gamma_i)k_2 \tag{17}$$

The laser heat source is modeled as a three-dimensional conical Gaussian heat source with a Gaussian interfacial energy distribution, which is expressed as follows [4]:

$$q(r,z) = \frac{9\eta Q e^3}{\pi(e^3-1)H(t)\left(r_e^2 + r_e r_i + r_i^2\right)}\exp\left(-\frac{\partial r^2}{r_0^2(z)}\right) \tag{18}$$

where Q, η, r_e, r_i, r, $r_0(z)$ and H(t) denote laser power, the absorption coefficient of metal to laser energy, Effective heat source radius of upper and lower surfaces, effective radius of laser beam from the target, and depth of heat source action, respectively, of which we set r_e to 0. 5 mm and r_i to 0.2 mm. $r_0(z)$ in equation (18). can be expressed as:

$$r_0(z) = r_e - (r_e - r_i)\frac{z_e - z}{H(t)} \tag{19}$$

The radius r in equation (18). can be expressed as:

$$r = \sqrt{(x-u_0t)^2 + y^2} \tag{20}$$

where t is the weld time and u_0 is the welding speed.

2.2. Boundary conditions

As shown in Fig. 2, the upper and lower parts are gas phase, and the left and right sides of the middle part are titanium alloy and aluminum alloy respectively. Set the upper surface EFGH as the velocity inlet, set the side of the gas phase as the pressure outlet, set the side and bottom of the two metallic materials as the ambient temperature and wall boundary, and the intersection of the two is set as the monitoring surface. The wall boundary considers only the energy loss due to convection and radiation, which can be expressed as [21]:

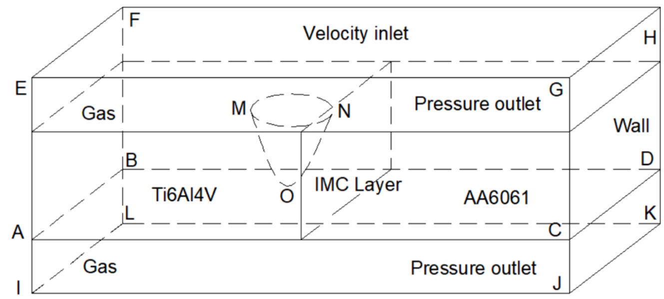

Fig. 2. Welding schematic diagram.

$$K\frac{\partial T}{\partial n} = h_A(T - T_0) - \sigma\varepsilon\left(T^4 - T_0^4\right) \tag{21}$$

where T, T_0, σ, ε and h_A denote the metal temperature, ambient temperature, Stefan-Boltzmann constant, surface emissivity, and convective heat transfer coefficient, respectively. The laser heat source acts on the surface of the workpiece to form a keyhole, and with time, the energy inside the keyhole reaches dynamic equilibrium, whose energy balance equation can be expressed as [21]:

$$K\frac{\partial T}{\partial n} = q - h_A(T - T_0) - \sigma\varepsilon\left(T^4 - T_0^4\right) - WH_{lg} \tag{22}$$

where q, WandH_{lg} denote the laser heat flux, evaporation rate, and latent heat of evaporation, respectively.

2.3. Numerical implementations

A three-dimensional model was developed using a fluid dynamics software named Fluent, and this mathematical model was discretized using the finite volume method (FVM). The material properties, boundary conditions and source terms (equations. (6), (8) and (22)) were defined by user-defined functions (UDFs), and the control equations were solved using the PISO (implicit pressure solver based on operator splitting) pressure–velocity coupling method. Table 1 shows the chemical composition of titanium alloy and aluminum alloy, Table 2 shows their thermophysical property parameters, and Table 3 shows the laser welding and offset parameters. During the welding process, the internal temperature of the molten pool rises sharply, and the thermophysical properties of the materials will change with the temperature. To obtain more accurate results, it is necessary to know the thermophysical properties of the materials at different temperatures. The density, thermal conductivity, specific heat and other parameters of the two materials are calculated by using JMatPro software. The length, width and thickness of the calculation field, are 10 mm, 10 mm and 4 mm, which can be divided into three parts: upper, middle and lower parts, which are gas phase with a height of 1 mm, metal phase in the middle, titanium material on the left and aluminum material on the right. The calculation time step is 1×10^{-5} s. To explore the relationship between the Ti-Al interface layer and IMC layer on the right side of the molten pool, the mass transfer module in Fluent was used for simulation, and the content of various substances in the Ti-Al interface layer was detected by probe method. The IMC produced by Ti-Al laser welding included TiAl3, TiAl2, TiAl3, etc., and the Ti-Al compound containing 20 %-80 % titanium was selected as IMC.

Table 2
Physical properties of Ti6Al4V and AA6061.

Parameters	Ti6Al4	AA6061
Density (kg/m^3)	4430	2720
Dynamic viscosity (kg/m·s)	0.005	0.0045
Specific heat (J/kg·K)	872	1020
Latent heat of fusion (J/kg)	3.6×10^5	4.92×10^5
Latent heat of evaporation (J/kg)	9.0×10^6	1.05×10^7
Thermal conductivity (W/m·K)	32.5	90
Liquidus temperature (K)	1933	872
Solidus temperature (K)	1877	812
Evaporation temperature (K)	3820	2790
Thermal expansion coefficient (K^{-1})	4.4×10^{-5}	2.38×10^{-5}
Surface tension (N/m)	1.68	0.82
Surface tension gradient (N/m·K)	-4.1×10^{-4}	-1.55×10^{-4}
Radiation emissivity	0.4	0.3

Table 3
Laser Welding and Offset Parameters.

Power (W)	Welding speed (m/s)	The offset distance (mm)	Heat input (J/mm^2)
1550	0.041	0.3	30
1550	0.041	0.3	38
1550	0.041	0.5	38
1550	0.041	0.1	38
1550	0.041	0	38
1550	0.055	0.3	38
1550	0.031	0.3	38
1300	0.041	0.3	38
1700	0.041	0.3	38

3. Results and discussion

3.1. Model validation

The simulation results show that when the parameter offset distance is 0.3 mm, the laser power is 1550 W and the welding speed is 2.5 m/min, the maximum welding width of the upper and lower parts of the molten pool is 1.57 mm and 1.35 mm respectively, and the minimum welding width of the middle part is 0.55 mm. Compared with the experimental results of Casalino et al.[22], as shown in Table 4, the shape and size of the cross section of the simulated molten pool are in good agreement with the experimental results, and the overall error is about 6 %.

Fig. 3 shows the comparison between our simulation profile and the experimental results of Casalino et al. [22]. The red dotted line and blue dotted line in Fig. 3 (b) show the experimental welding profile and

Table 4
Comparison of experimental and simulation results.

	Experimental results(mm)	Simulation results(mm)	Percentage Error(%)
Upper surface melting width	1.52	1.57	3.29
Lower surface melting width	1.33	1.35	1.50
Minimum melting width	0.82	0.79	3.66
Average melting width	1.09	1.16	6.42

Table 1
Chemical composition of Ti6Al4V and AA6061 (wt.%).

AA6061	Mg	Si	Fe	Cu	Mn	Ti	Zn	Cr	Al
	0.45	0.47	0.21	0.05	0.07	0.06	0.09	0.03	Base
Ti6AlV	V	Si	Fe	O	N	Ti	H	C	Al
	4.10	0.15	0.3	0.20	0.05	Base	0.01	0.10	5.82

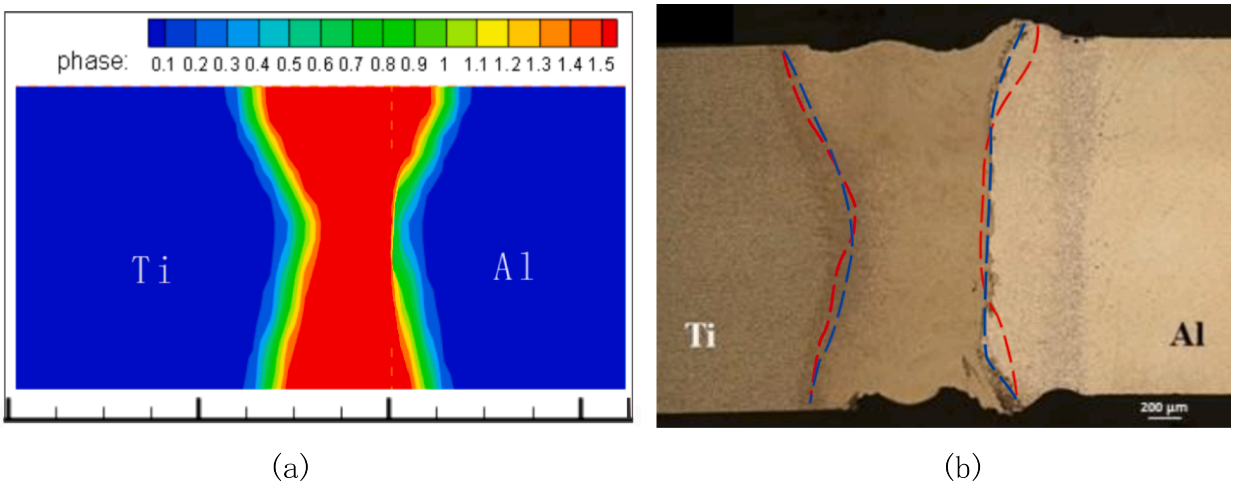

Fig. 3. Cross-sectional morphology diagram of the melt pool: (a) simulated melt pool cross-sectional diagram, (b) experimental cross-sectional diagram of titanium-aluminum offset welding with two lines inside the diagram (red dashed line is the simulated melt pool contour line and blue dashed line is the experimental melt pool contour line) [22],

simulated welding profile respectively, and there are some errors between them, with the error on titanium side being small and that on aluminum side being slightly larger, which may be due to the fact that the 35 μm gap between titanium and aluminum is neglected in the assumption.

3.2. Fluid changes in the weld pool

Fig. 4 shows the distribution of welding temperature field and velocity field in the YZ section under conditions of P = 1550 W and V = 0.041 m/s. Laser energy acts on the titanium surface. With the input of energy, the titanium surface gradually melts, forming a small molten pool at the border of laser and material. The flow velocity at the bottom of the molten pool is faster than that at the two sides, as shown in Fig. 4 (a). The molten pool gradually expanded to the boundary surface of titanium and aluminum materials, and melted the aluminum materials by heat conduction. With the fusion and reaction of the melted titanium and aluminum materials, a thin IMC (intermetallic compound) layer was

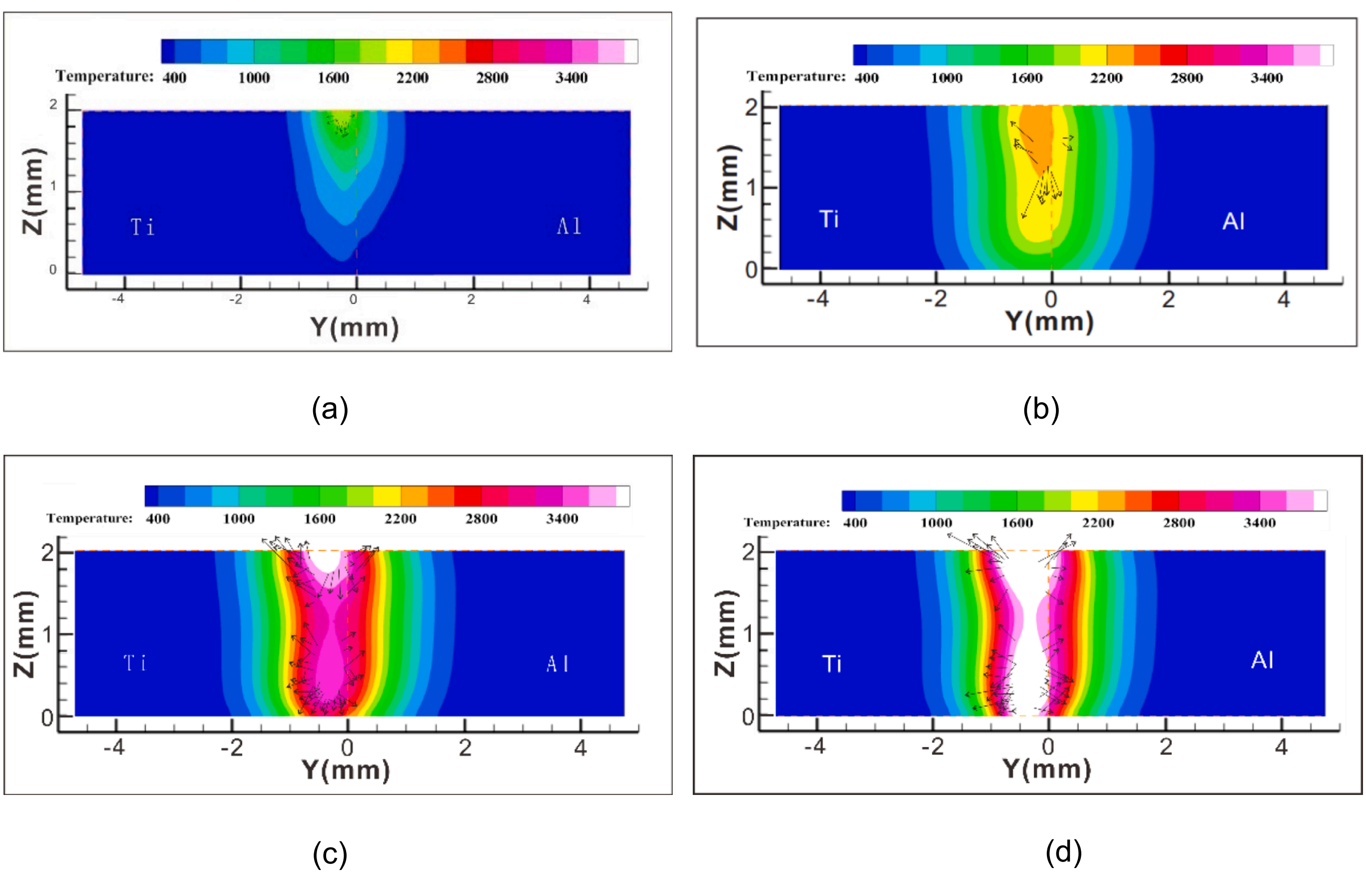

Fig. 4. Distribution of temperature field and velocity field in YZ section with time (P = 1550 W, V = 0.041 m/s): (a) = 3.50 ms, (b) = 7.60 ms, (c) = 15.20 ms and (d) = 20.50 ms.

formed at the boundary. However, the IMC affects the performance of the welded joint to a certain extent [23]. Reducing its thickness can significantly increase the quality of the joint. The closer to the laser beam, the higher the temperature of the molten pool. When the temperature of liquid metal is higher than the boiling point of the metal, it will cause severe evaporation. The evaporation of metal will produce a lot of plasma, which will gather on the surface of the keyhole. It will absorb laser energy and will hinder the absorption of laser energy by keyhole wall. Under the action of recoil pressure, the depth of the molten pool increases continuously, and the surface tension near the keyhole plays an obstacle role. When the width of the molten pool tends to be stable, more laser energy is transmitted to the bottom of the molten pool, which makes the molten pool expand around by heat conduction. Due to the latent heat of phase change, an X-shaped welding contour is formed at the initial position of laser action, which is similar to the results of Casalino et al.[11]. The uneven temperature distribution in the whole molten pool makes the surface tension of the liquid metal having a gradient distribution, which leads to Marangoni force. From Fig. 4(c), it can be seen that the liquid metal flows rapidly at the top and bottom of the molten pool, and the liquid metal moves upward from the middle of the molten pool, forming a dynamic thermal cycle at the top of the molten pool. The more uneven temperature distribution in the molten pool and the faster the flow rate, the stronger Marangoni convection effect. From Fig. 4(c) and Fig. 4(d), the velocity density distribution on the left and right sides is much lower than that on the upper and lower sides, which leads to the appearance distribution of the molten pool with large two ends and small middle. When the time is 20.50 ms, the maximum flow rate in the upper part of the molten pool is 25.3 m/s, the maximum flow rate in the lower part is 23.4 m/s, and the melting width in the upper part is slightly larger than that in the lower part.

Fig. 5 is a top view of the workpiece, from which it can be seen that the flow of molten metal around the keyhole changes, and the molten metal is forced to flow outward from the center. The laser beam advances along the X-axis direction, and the molten metal flows from front to back along the wall surface of the keyhole, and gradually solidifies at the tail of the keyhole to form a weld. It can be seen that there is an obvious temperature gradient change at the interface between titanium and aluminum, and the velocity vectors on both sides of the keyhole are relatively dense, with a tendency to expand around.

Fig. 6 shows the temperature field and velocity field of the middle surface of the molten pool at t = 38 ms. The laser beam advances along the X axis, and the temperature gradient in the front side of the molten pool is obvious. The solidification of the molten metal in the rear side and a large amount of input of the molten metal in the front side along the wall of the keyhole make the rear side have a wider temperature gradient. It can be observed that the velocity vectors of the upper and lower parts of the molten pool are relatively dense, and the main reason

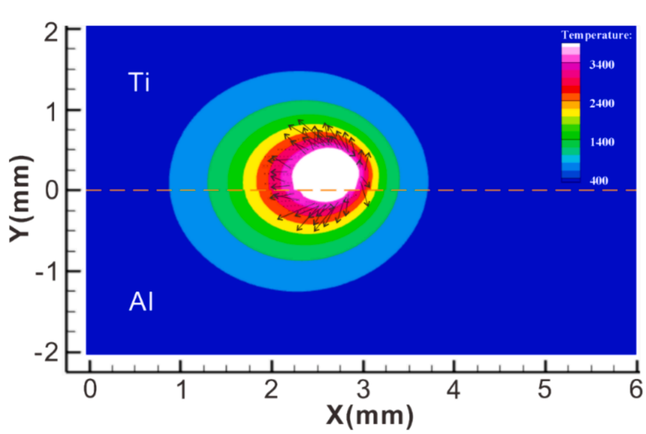

Fig. 5. Temperature and velocity fields of the top view of laser welding at time: t = 20.50 ms (P = 1550 W, V = 0.041 m/s).

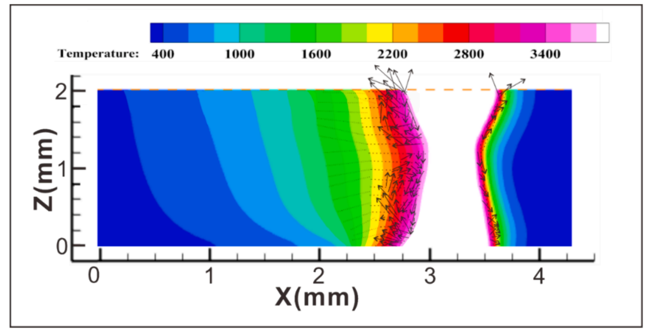

Fig. 6. Temperature and velocity fields in the middle surface of the molten pool at time t = 38 ms(P = 1550 W, V = 0.041 m/s).

for this vector distribution is that the workpiece is influenced by the recoil pressure of the molten metal surface. The middle part of the melt flows in two directions respectively, one part flows upward and the other part flows downward. A slight reflux trend can be observed in the upper part of the solution, and a small thermal cycle is formed inside the fluid. These phenomena are all caused by the Marangoni convection effect [24]. The laser beam irradiated on the bottom of the keyhole will make the molten pool flow downward under the influence of recoil pressure, surface tension, gravity, hydrostatic pressure, etc., and the inside of the molten pool will be squeezed and flow backward. In the welding process, the increase of keyhole surface temperature will increase the recoil pressure, and at the same time, the surface tension will decrease. Because the upper part of the molten pool receives a lot of laser energy, its temperature will become very high, and the surface tension coefficient will become very small. The small surface tension and high recoil pressure will slow down the liquid flow on the top surface of the molten pool, thus inhibiting the backflow of the upper part of the molten pool. Because the bottom of the keyhole is close to the focal plane, it absorbs a large amount of laser energy, resulting in a larger recoil pressure. The recoil pressure in the middle of the molten pool is small, the surface tension coefficient is large, and the surface of the molten pool flows downward, thus generating a clockwise vortex and forming a local thermal cycle, which is similar to the research results of Lin et al [25].

3.3. Effect of laser power, welding speed and offset

The changes in welding speed, welding power and offset will affect the welding performance of welded joints to varying degrees. In order to improve the welding performance, the most common method is to reduce the thickness of IMC layer by changing the welding process parameters. Assuming that the thickness and morphology of IMC layer are positively related to the thickness and morphology of Ti-Al interface layer on the right side of molten pool. To explore this relationship, the welding temperature field under four different parameters was studied, as shown in Fig. 7, which is the welding temperature field of YZ section under different process parameters at t = 20.50 ms, The liquefaction temperature of titanium alloy used in the research is about 2000 K and that of aluminum alloy is 873 K. Considering the transport and diffusion of materials, the IMC layer must be between the Ti-Al interface layer with the temperature of 2000 K to 3500 K Fig. 7(a) is the control group set up in the simulation.

As shown in Fig. 7(b), when the welding speed is increased, the weld width and keyhole depth are correspondingly reduced. With the increase of welding speed, the energy received by the workpiece in unit time is reduced, and the penetration rate of laser to materials is reduced. Properly increasing the welding speed can reduce the growth of Ti-Al interface layer, reduce the thickness of IMC layer and improve the welding performance of welded joints, which is similar to the research results of Lee et al [26]. By comparing Fig. 7 (a) and (c), increasing the welding offset while remain other process parameters unchanged, the

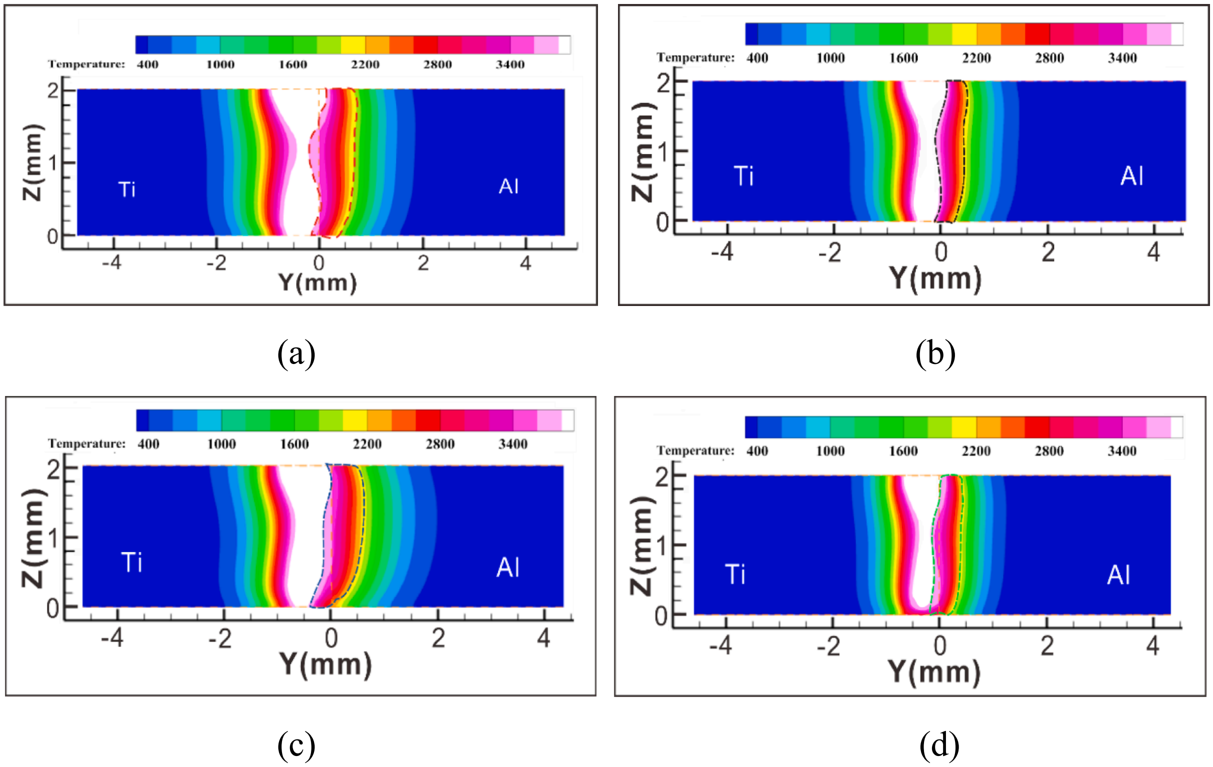

Fig. 7. Temperature field in the middle surface of the molten pool at time t = 20.50 ms, The red, black, blue and green dotted lines respectively indicate the intermetallic transition layers at different welding speeds, offsets and powers: (a)= (P = 1550 W, V = 0.041 m/s, O = 0.3 mm), (b)= (P = 1550 W, V = 0.055 m/s, O = 0.3 mm), (c)= (P = 1550 W, V = 0.041 m/s, O = 0.5 mm), (d)= (P = 1300 W, V = 0.041 m/s, O = 0.3 mm).

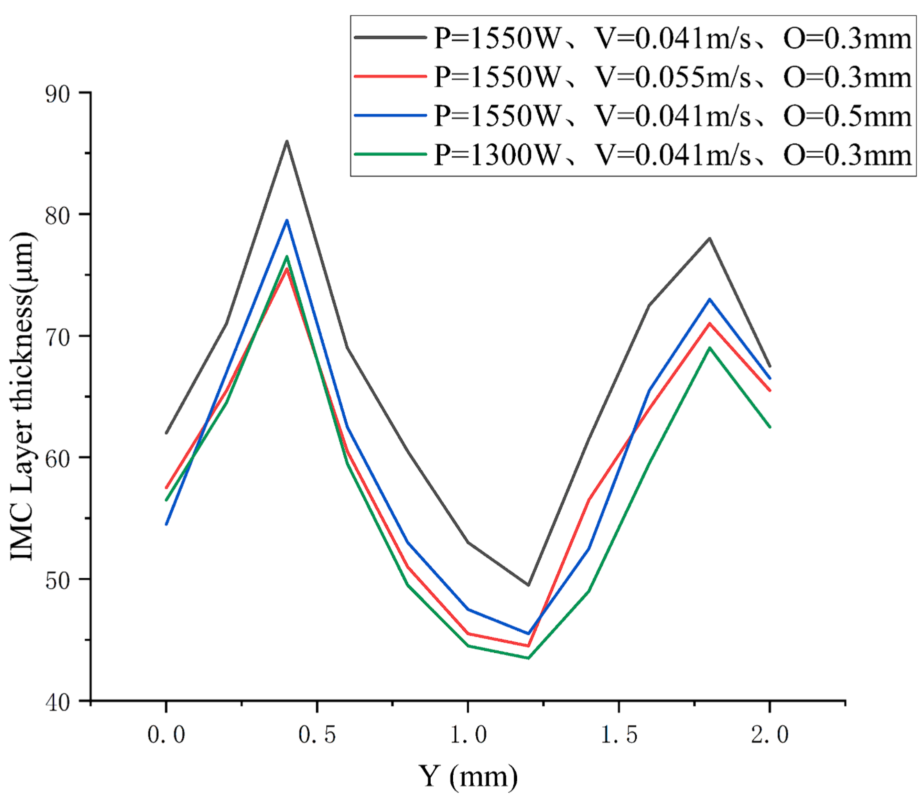

Fig. 8. The thickness of IMC layer under different welding speeds, offset, and power at time t = 20.50 ms.

width of the molten pool near the aluminum side will decrease. The melting point difference between the two metal materials is about 1000 K. When the molten pool expands to the interface between titanium and aluminum, the aluminum alloy will instantly liquefy due to higher temperature and enter the left molten pool, and the two liquid metals will mix and react with each other to form IMC. In Fig. 7(c), the increase of the offset within a certain range, will reduce the thickness of the titanium-aluminum interface. From Fig. 7(d), we can see that the shape of the Ti-Al interface layer formed by reducing laser power is similar to increasing welding speed.

From Fig. 8, it is observed that the thickness of IMC layer under different welding speeds, offset, and laser power at t = 20.50 ms, the red line segment is the control group, and there is a convex trend at the height of 0.4 mm and 1.8 mm. Increasing the welding speed, offset, and laser power within a certain range can reduce the thickness of IMC layer, thus improving the performance of welded joints. From the observation of Fig. 8, it can be seen that increasing the welding speed and reducing the laser power have a slightly greater inhibitory effect on the formation of IMC layer than increasing the offset. Under different power and speed, the depth of keyhole will change constantly.

Fig. 9 shows the distribution of IMC layer at the interface of titanium and aluminum when t = 20.50 ms, P = 1550 W, V = 0.041 m/s and O = 0.3 mm. Red represents titanium, blue represents aluminum, and the phase on the right represents different titanium content. In Fig. 9, the IMC layer in the middle is thinner than upper and lower sides. Compared with Fig. 7(a), it is concluded that IMC is easier to form in places with dense temperature gradients.

Fig. 10 shows the keyhole morphology under different welding parameters at time t = 35.50 ms. The red area represents the metal substance and the white area represents the air. The maximum keyhole depth at different times is measured, and the change of keyhole depth with time in the welding process is obtained. As shown in Fig. 11 the red line is the control group, the black line segment represents the change of keyhole depth under high laser speed, and the green line segment represents low power. The depth of the keyhole fluctuates with time, in which the black line segment fluctuates less and the green line segment fluctuates more. Due to the recoil pressure, the keyhole depth increases continuously, and the surface tension of keyhole hinders the recoil pressure. When the surface tension is large enough, keyhole collapse will occur, small bubbles will form in the molten pool, and the keyhole depth will decrease. With the continuous input of laser energy, the recoil pressure will gradually increase. The fluctuation of keyhole depth is the result of the interaction between recoil pressure and surface tension, and the dynamic balance between them keeps the key Kong Wei in a certain shape. The results show that under the action of high power and high welding speed, the width of the upper surface of keyhole is larger, which reduces the probability of keyhole collapse and makes keyhole more stable.

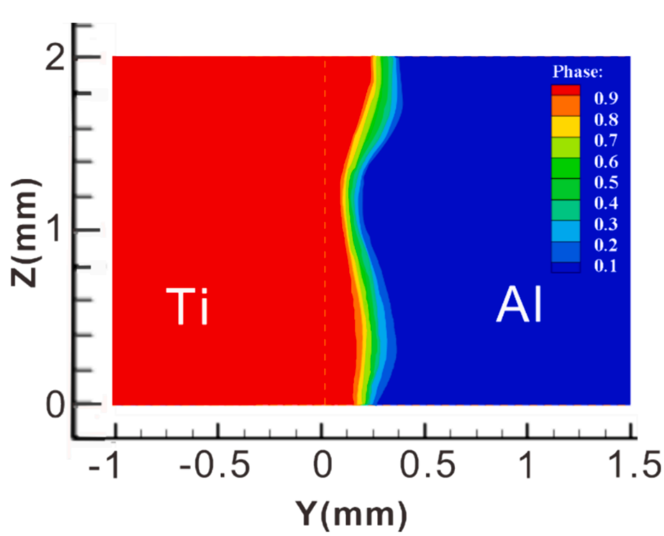

Fig. 9. Mass transfer distribution at the interface of titanium and aluminum at time t = 20.50 ms, P = 1550 W, V = 0.041 m/s and O = 0.3 mm.

3.4. Analysis of internal changes of molten pool by dimensionless number

The flow and heat transfer in the molten pool are affected by various driving forces, and the analysis of dimensionless numbers plays an important role in understanding the internal forces in the molten pool. The width of molten pool is unevenly distributed. To better reflect the relationship between dimensionless number and molten width, the width of molten pool is now converted into dimensionless width W_A, and the variable is unitless. The flow state of the fluid in the molten pool can be characterized by Reynolds number as shown in equation (23).

$$\mathrm{Re} = \frac{\rho U L_a}{\mu} \tag{23}$$

where U, L_a represent flow velocity and characteristic length. Fig. 12(a) shows the relationship between dimensionless width and Reynolds number under different parameters. It can be seen from the figure that Reynolds number is higher at higher welding power and lower welding speed, and the offset has little effect on Reynolds number, and the flow state of molten pool is laminar. Peclet number can be used to characterize the heat transfer mechanism in the molten pool during welding, and it is expressed by equation (24).

$$Pe = \frac{\rho U L_a C_P}{K} \tag{24}$$

Peclet number is the ratio of convection heat transfer rate to diffusion heat transfer rate in the molten pool. As shown in Fig. 12(b), the Peclet number is greater than 1, which indicates that the main heat transfer mechanism in the molten pool is convection heat transfer. Under the condition of higher welding power and lower welding speed, the convection in the molten pool is more intense when the Peclet number is higher. The driving force inside the molten pool is mainly dominated by the surface tension and buoyancy, and through the equation (25).

$$\mathrm{R}_{s/b} = \frac{Ma}{Gr} \tag{25}$$

Understand the action mode of driving force, where *Ma* represent Marangoni number and *Gr* represent Grashof number. As shown in Fig. 12(c), it is obviously $\mathrm{R}_{s/b}$ greater than 1, which indicates that the interior of the molten pool is subjected to a relatively strong Marangoni force, and Marangoni convection dominates.

3.5. Mass transfer

Species transportation includes fluid convection and material diffusion, and the concentration difference between titanium and aluminum will cause diffusion. Offset welding is essentially a process of material heat conduction. The diffusion studied in this paper is a typical downhill diffusion. In the process of simulation calculation, the concentration distribution of substances is obtained by solving equation (9). In this paper, the transition layer with the concentration of titanium, aluminum and air ranging from 10 % to 70 % is defined. As shown in Fig. 13(a), it is the diffusion diagram of metal in the air. Under the input of laser energy, the metal material changes from solid to liquid. When the temperature in the keyhole reaches the boiling point of metal, a large amount of metal steam will be produced. The steam concentration on the surface of the material is about 10.3 % measured by the probe method in the post-processing software Tecplot. This shows that some metal materials have spilled into the air through steam. From Fig. 13(a), we can observe the mass transfer inside the keyhole. The formation of keyholes is an

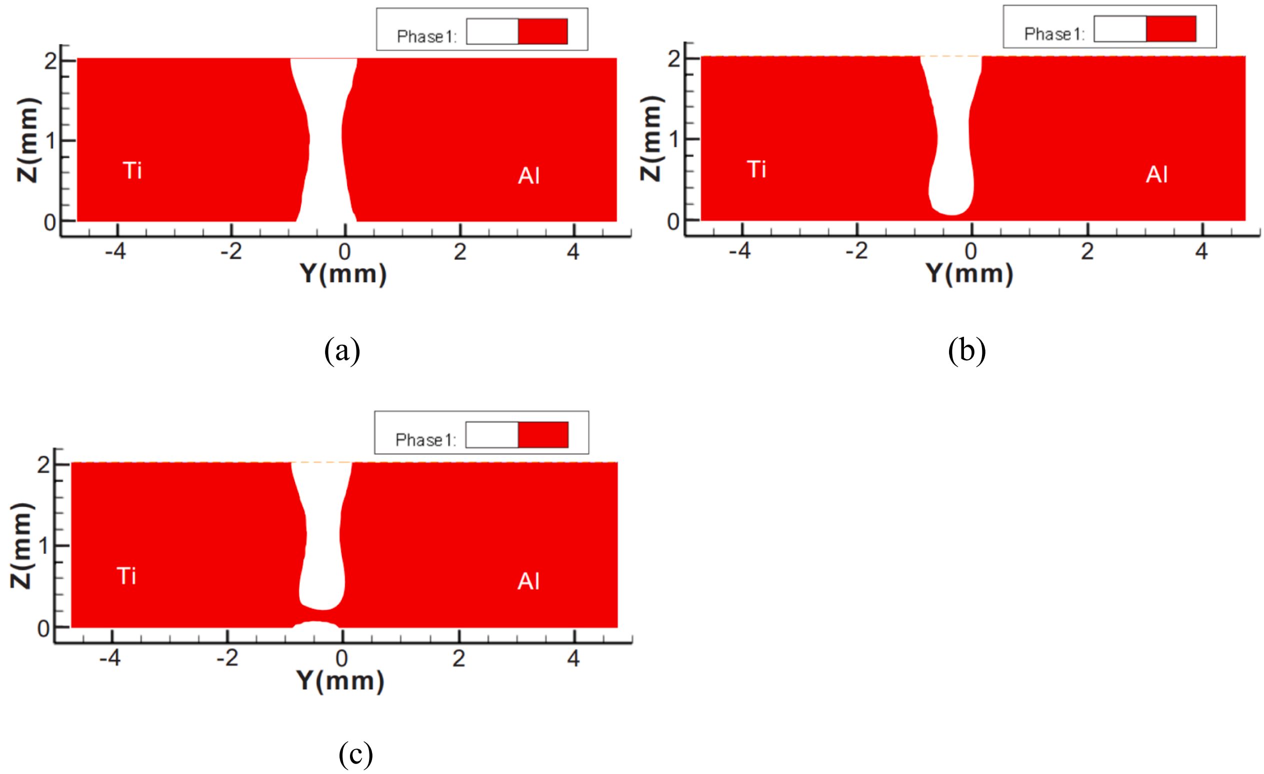

Fig. 10. The appearance of keyhole at time t = 35.50 ms:(a)= (P = 1550 W、V = 0.041 m/s、O = 0.3 mm)、(b)= (P = 1550 W、V = 0.055 m/s、O = 0.3 mm)、(c)= (P = 1300 W、V = 0.041 m/s、O = 0.3 mm).

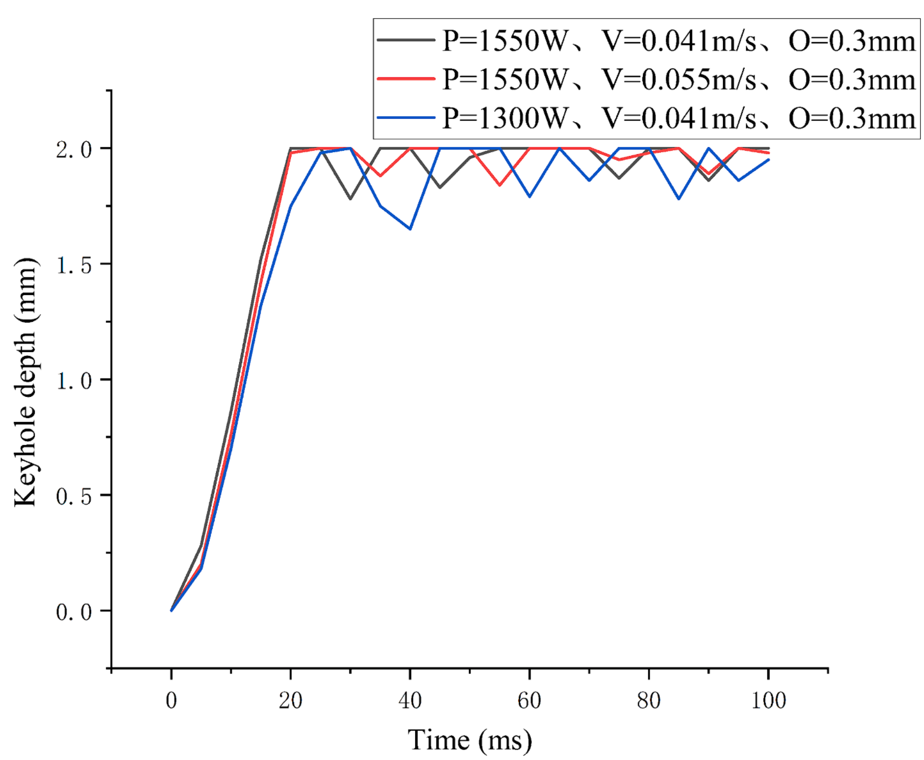

Fig. 11. Evolution curve of keyhole depth under different parameters.

important factor to promote mental transportation. During the process of keyhole expansion, pressure will be formed around the keyhole, which will promote the right transportation of titanium. Obviously, at t = 20.50 ms, the keyhole has penetrated the bottom of the metal. As shown in Fig. 13(b), there is a large amount of metal vapor inside and at the bottom of the keyhole. From the distribution of aluminum, there is little aluminum in the keyhole, and all of these vapors are generated by the evaporation of titanium. There are a lot of intermetallic compounds

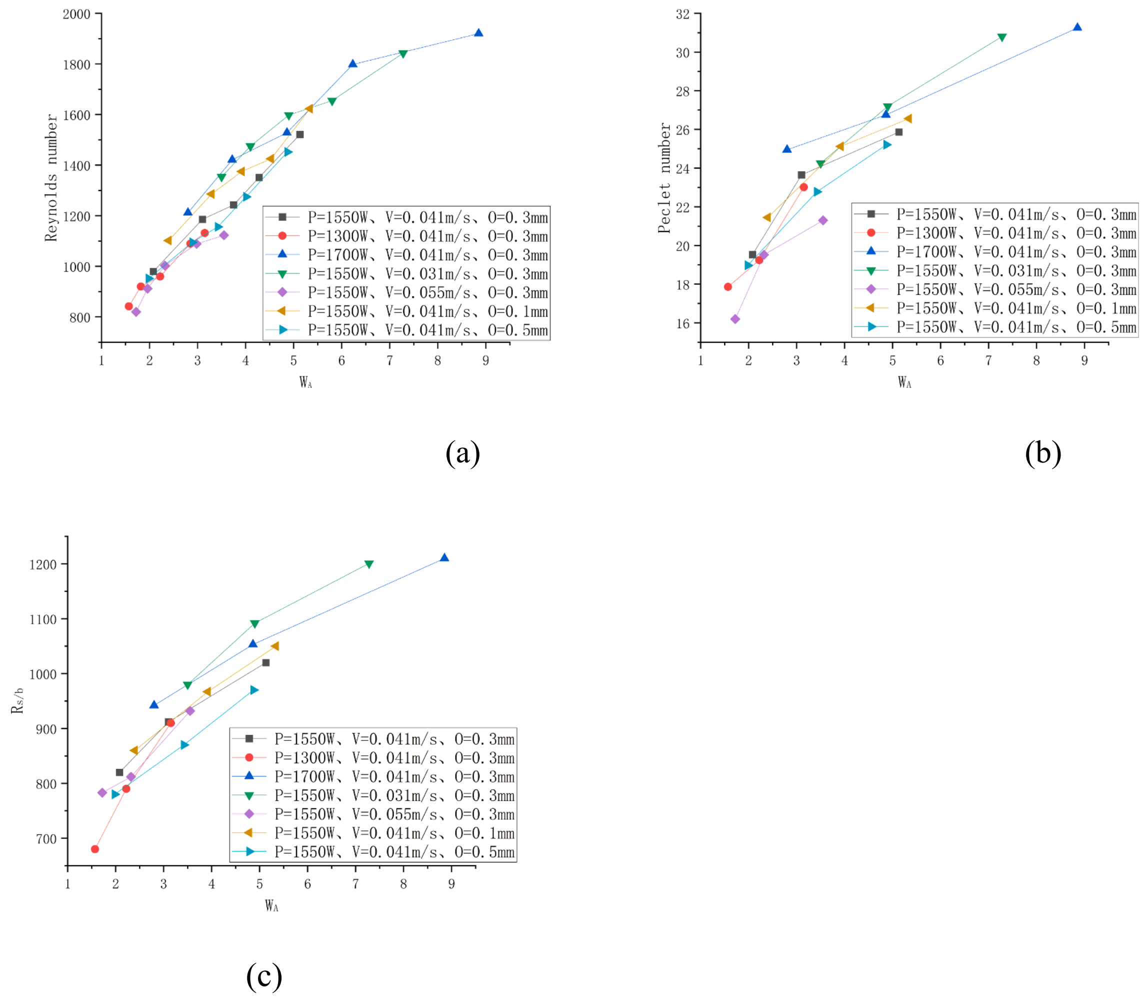

Fig. 12. (a) the relationship between dimensionless width and Reynolds number, (b) the relationship between dimensionless width and Peclet number, (c) the relationship between dimensionless width and $R_{s/b}$.

in the transition layer of titanium. From the weld to the transition layer, the concentration of titanium decreases, and the concentration of aluminum increases in turn. According to Fick's law, high concentration titanium diffuses to low concentration through the transition layer, and similarly, high concentration aluminum diffuses to the weld zone. In the process of molten pool flow, liquid aluminum diffuses into the weld zone by convection, and blends with liquid titanium in the transition layer, which shows that convection plays an important role in mass transfer between titanium and aluminum. In Fig. 13(b), obvious bulge and sag of the molten pool can be seen. When the recoil pressure and hydrostatic pressure generated by titanium vapor are greater than the surface tension in the keyhole, sag and sag will occur, and the sagging part is often larger than the bulging part due to gravity. And sag and bulge are more obvious on the aluminum side, because the melting point of aluminum is much lower than that of titanium, and the concentration of titanium on the aluminum side is very low, which is prone to convection and diffusion. In Fig. 13(b), it can be observed that the transition layer between titanium and aluminum has a growing trend of big at both ends and small in the middle, which is due to the effect of surface tension on the molten pool, and part of the surface tension is driven by concentration, which is called solute capillary effect, and part of the surface tension changes with temperature, which is called thermal capillary effect. Both of these effects belong to Marangoni effect. Under the influence of double effects, the transition layer is in a different state. The upper and lower surfaces of the keyhole absorb more laser energy, which leads to high temperature, dense temperature gradient, and strong thermal capillary effect, which intensifies the convection–diffusion trend of liquid titanium to the aluminum side and increases the thickness of the transition layer.

4. Conclusions

In this paper, the dynamics of the weld pool of AA6061 and Ti6Al4V under laser offset butt welding is studied, a three-dimensional butt numerical model is proposed, the influence of different parameters on the interior of the weld pool is studied, and the mass transportation between titanium and aluminum is calculated. The main conclusions of this study are as follows:

(1) When the laser power is 1550 W, the welding speed is 0.041 m/s, and the offset is O = 0.3 mm, the maximum flow rate in the molten pool is 28.6 m/s and the maximum melting width is 1.57 mm. The keyhole will penetrate the bottom of the material, and the hole depth will fluctuate due to collapse.

(2) The recoil pressure promotes the formation of keyhole, while the surface tension hinders the formation of keyhole. The flow of molten pool is controlled by Marangoni force and influenced by keyhole. The bigger and deeper the keyhole, the faster the flow rate of the molten pool and the more intense the flow.

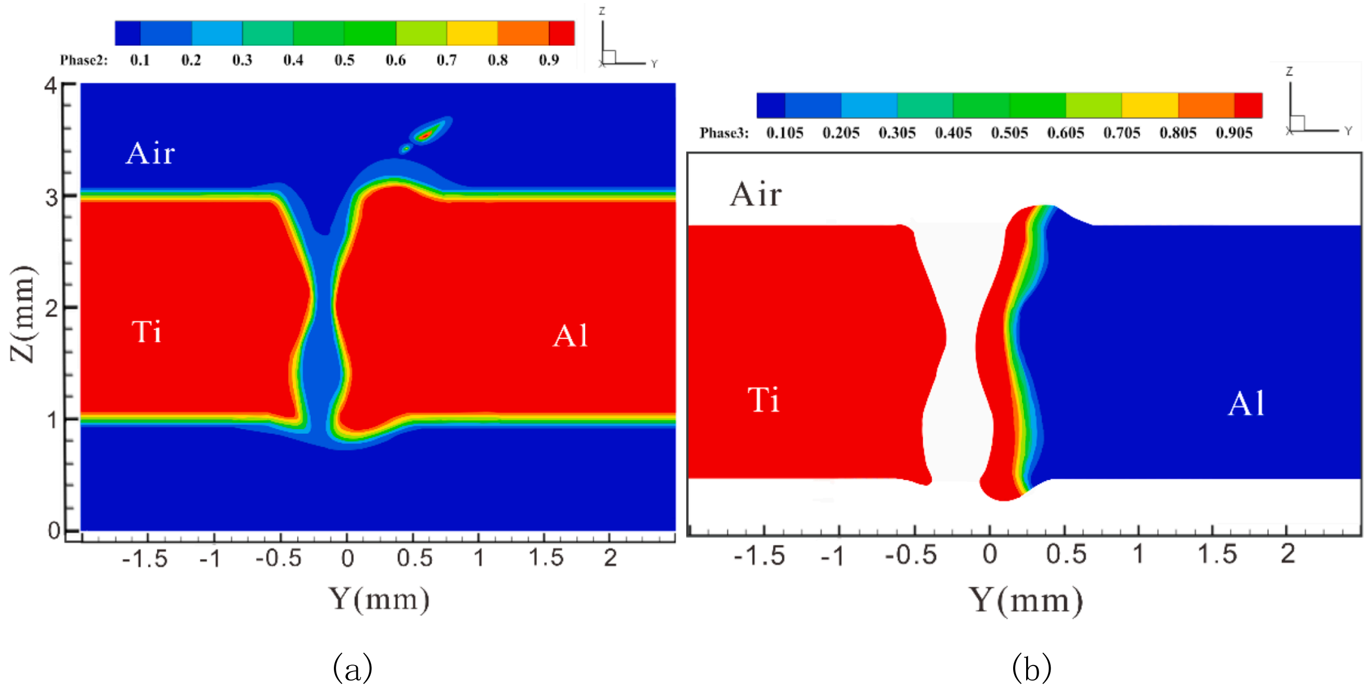

Fig. 13. (a) Mass transfer distribution of metal materials in the air at time t = 20.50 ms.(b) Mass transfer distribution between titanium and aluminum at time t = 20.50 ms.

(3) In the welding process, the flow state in the molten pool is mainly laminar flow, and its heat transfer mechanism is mainly convective heat transfer, with surface tension as the main driving force.
(4) Mass transportation is inseparable from convection and diffusion. Different welding parameters will produce intermetallic transition layers with different thicknesses. Increasing welding speed and decreasing laser power within a certain range can inhibit the growth of intermetallic transition layers. Precise control of offset is the key to reducing intermetallic compounds.

Porosity is an important factor that affects the welding quality, but this work has not made an in-depth analysis of porosity. To improve the overall welding effect, the formation of porosity and the factors affecting porosity can be analyzed from the aspects of internal force and flow in the molten pool. It is also possible to consider the behavior of molten pool caused by increasing the heat affected zone and the change of porosity when using oscillating laser or double-beam laser welding.

CRediT authorship contribution statement

Jinbo Yu: Conceptualization, Data curation, Formal analysis, Investigation, Software, Writing – original draft. **Xigui Xie:** Data curation, Investigation, Resources, Supervision, Visualization, Writing – original draft. **Shanshi Huang:** Software, Validation, Visualization, Writing – review & editing. **Jianxi Zhou:** Funding acquisition, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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